

Chapter X

Project

Depth Estimation

- Imagine you have a color image



RGB Image

Depth Estimation

- Imagine you have a color image
- And you want to estimate the depth map



RGB Image

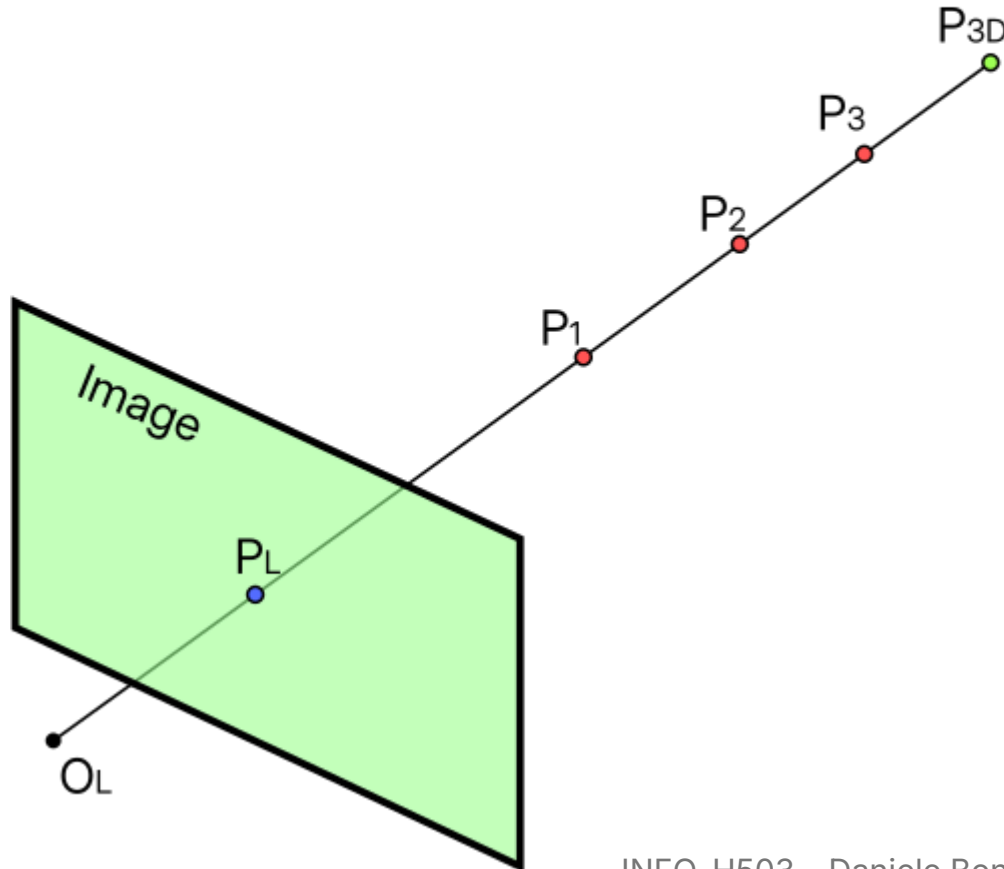


Depth Map

- How do you do ?

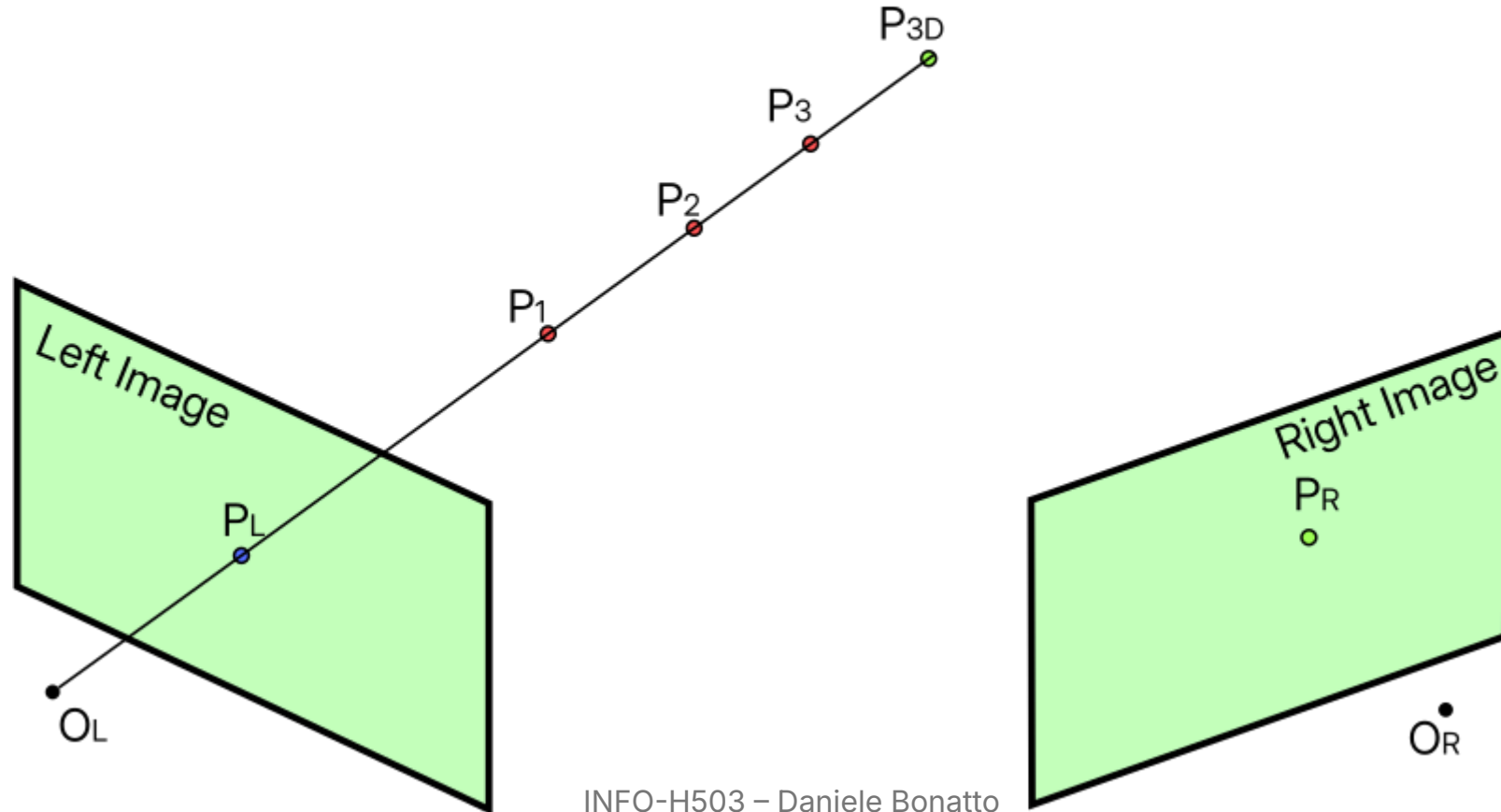
Depth Estimation

- With only one color image, it is difficult !
- The 3D point could be anywhere along a ray

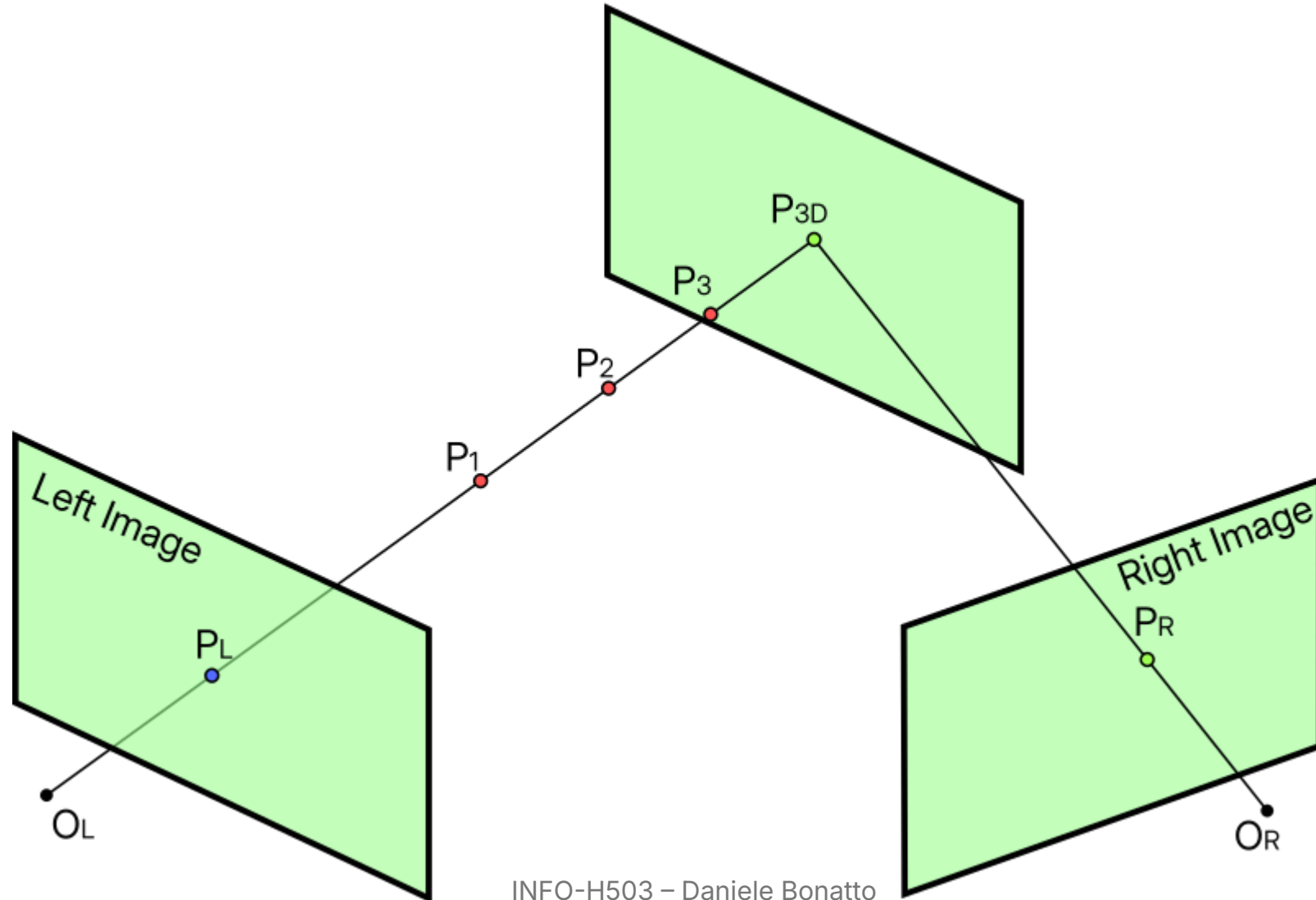


Depth Estimation

- Can we improve the situation with 2 images ?

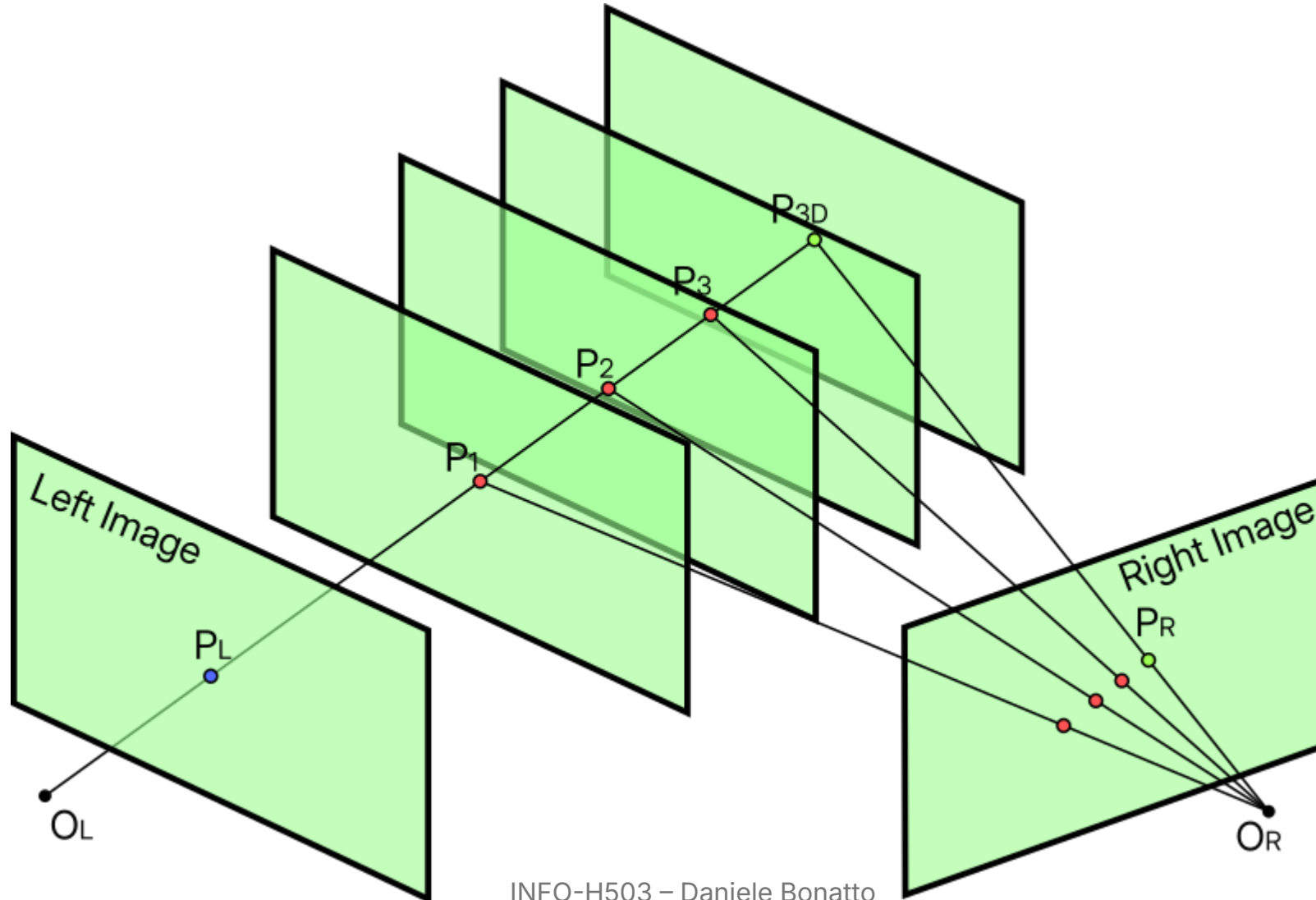


- If we move the left image at the 3D point.. When we *project* it to the right image, we should find a correspondance !



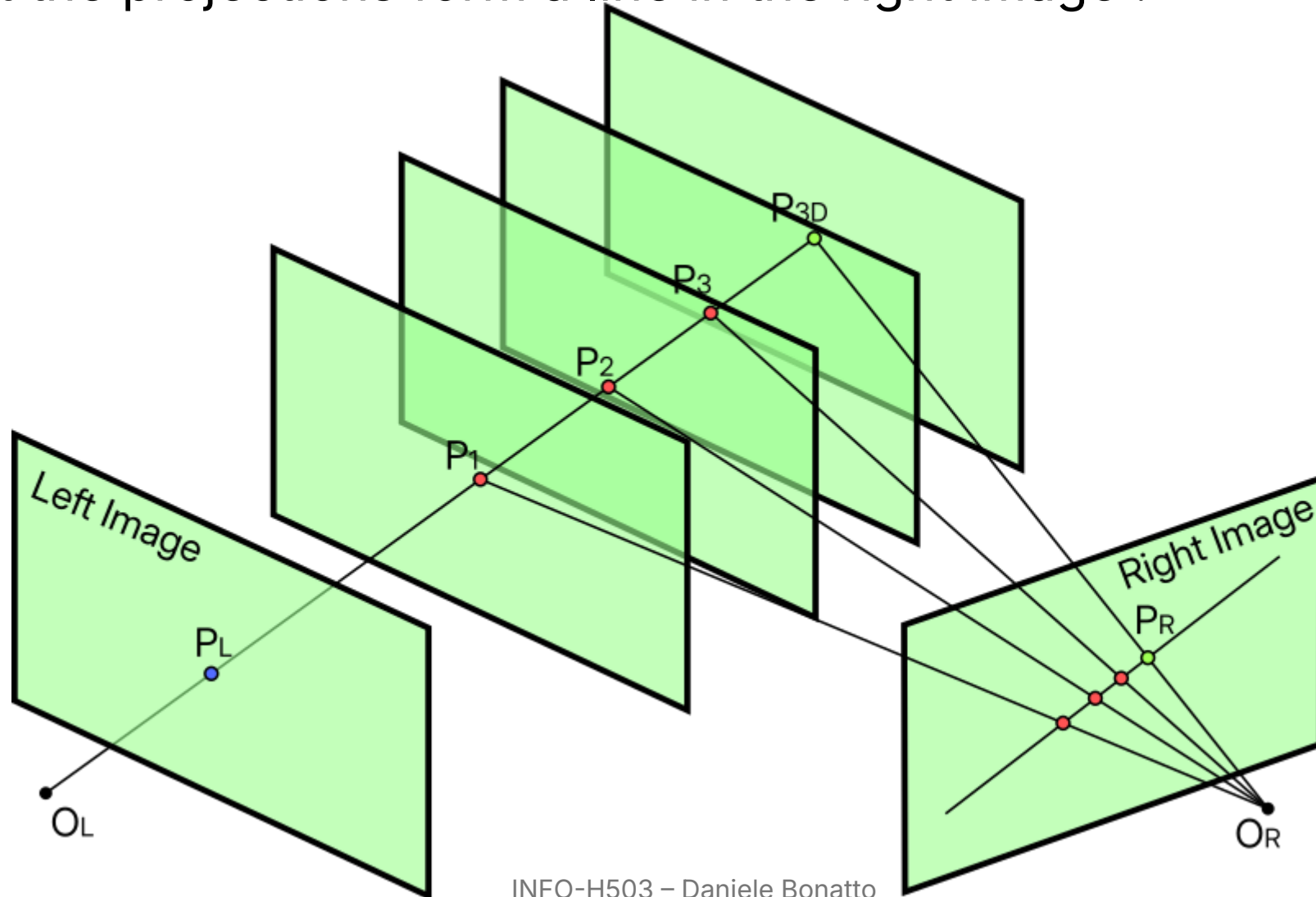
Depth Estimation – Plane Sweeping Algorithm

- But.. We don't know where to put our plane ☹



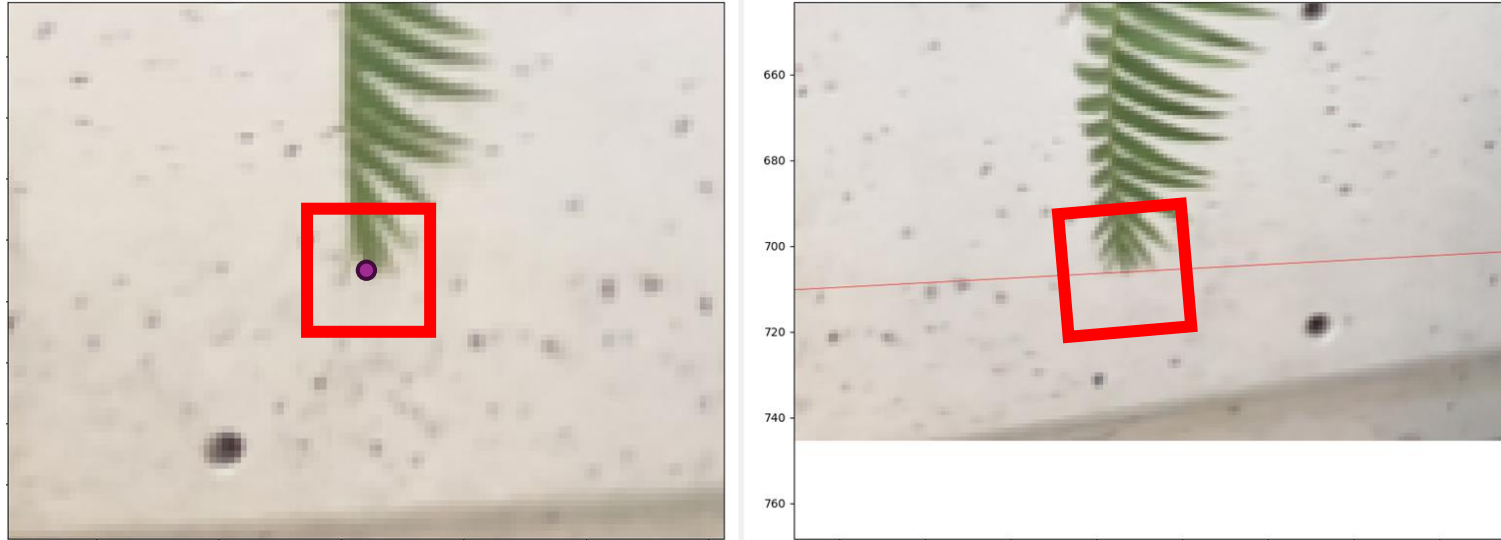
Depth Estimation – Plane Sweeping Algorithm

- We can try them all ! This is called Plane sweeping algorithm.
- Notice that the projections form a line in the right image !



Depth Estimation

- To increase accuracy, we can compare a small window around the left and right images

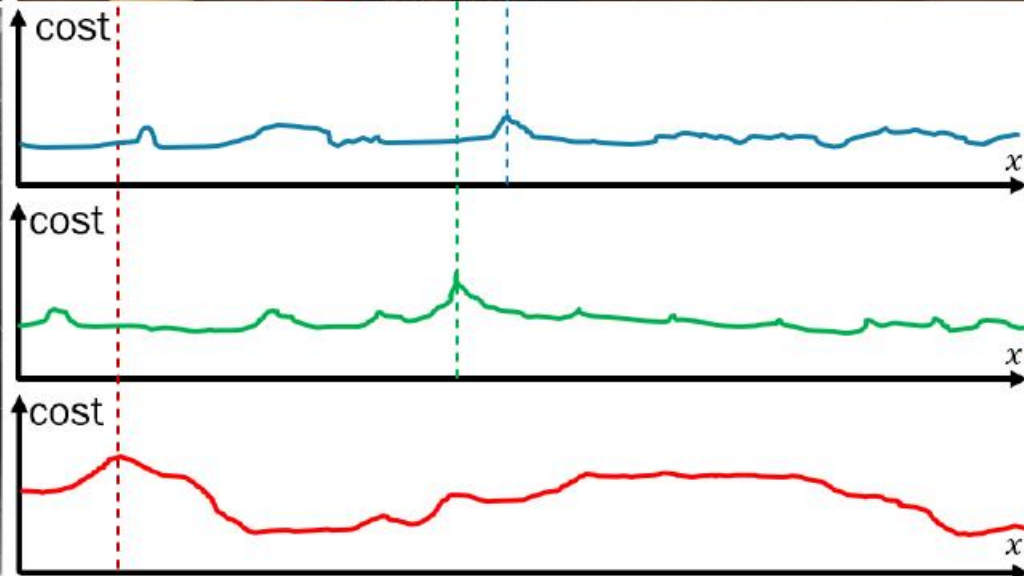
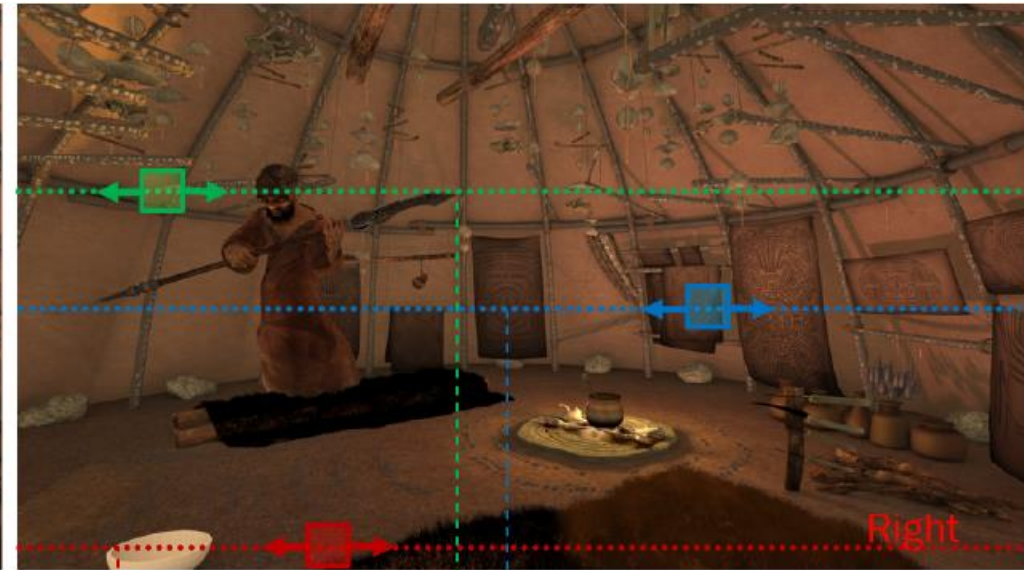


- We can use the absolute difference of all pixels inside the windows, then divide by the size of the windows

$$SAD(w_1, w_2) = \sum_{x,y} \frac{|w_1(x, y) - w_2(x, y)|}{\text{area}}$$

- This is called Sum of Absolute Difference (SAD)

Depth Estimation

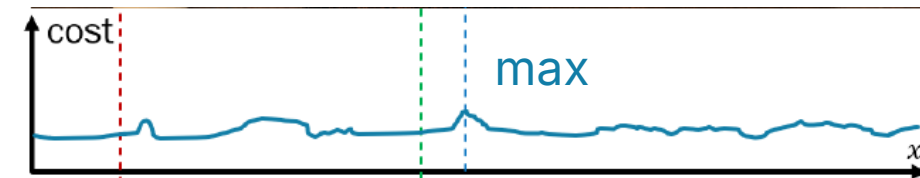


Depth Estimation

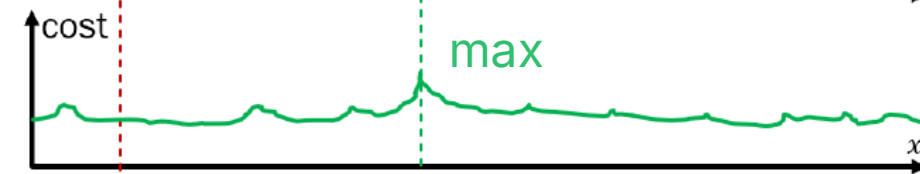
- If we keep the maximal cost for each pixel
- We get a pretty bad depth map ☹️



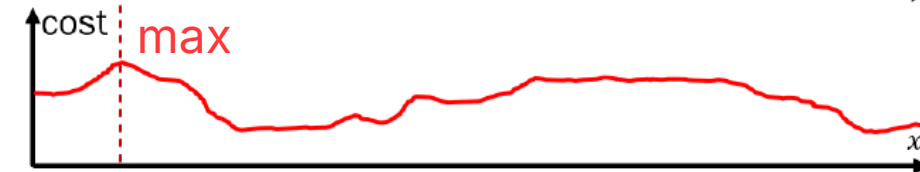
One pixel



One pixel



One pixel

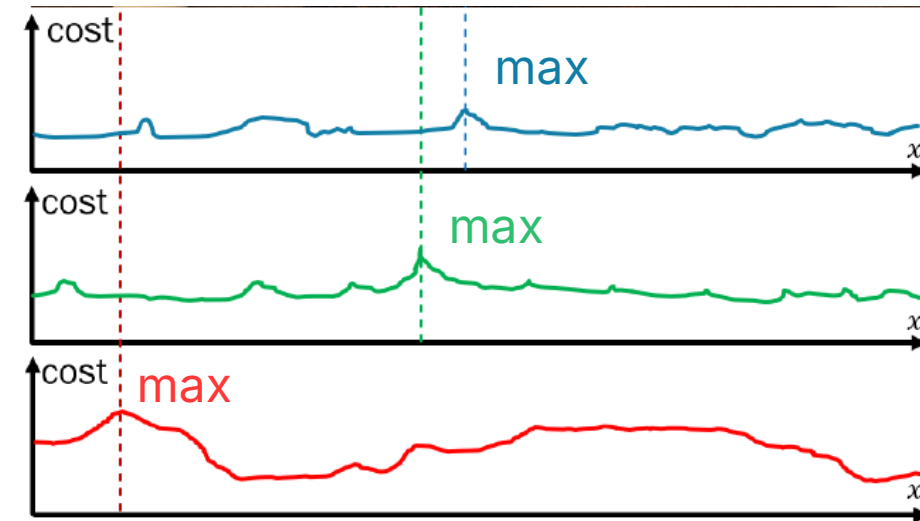




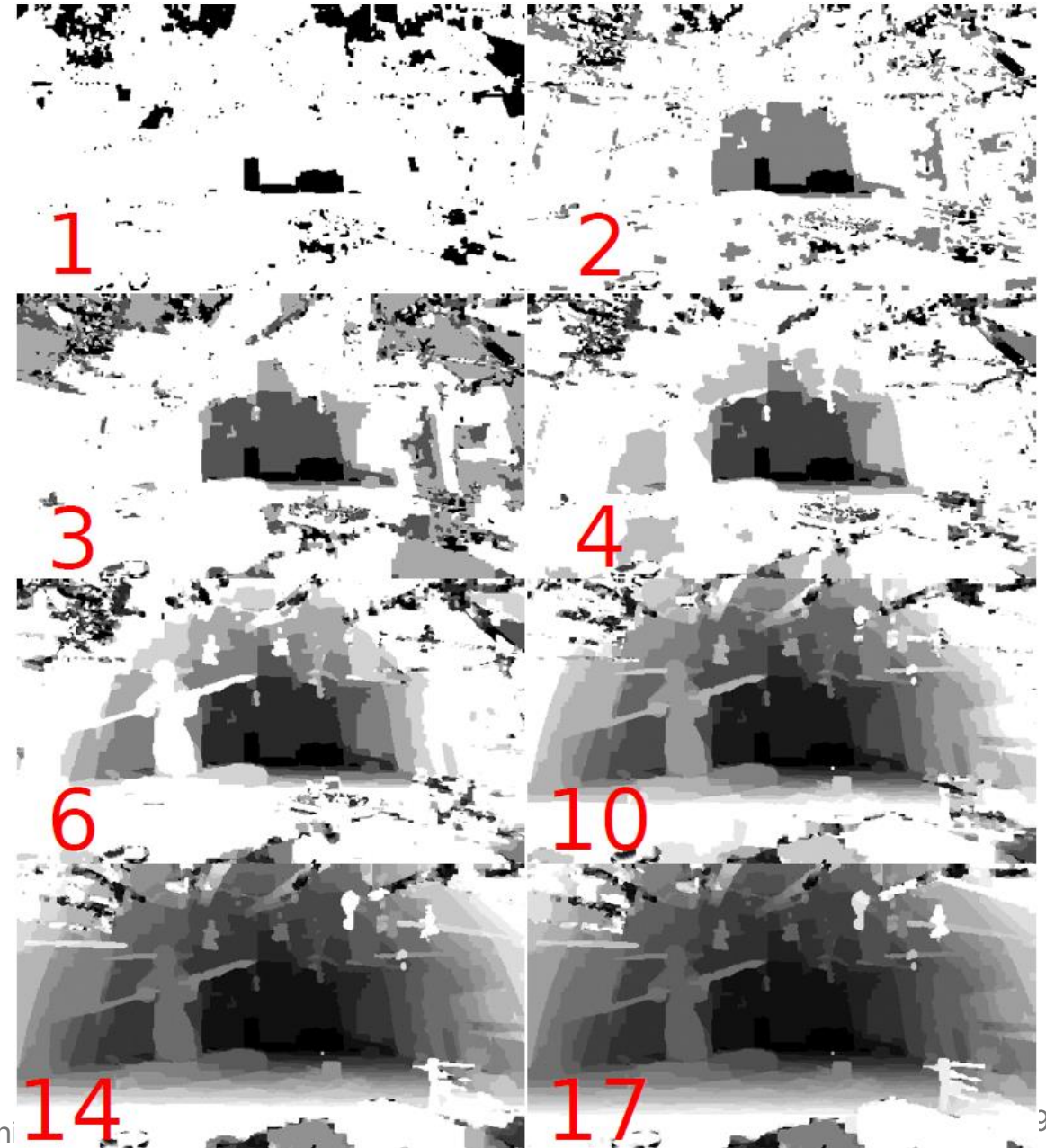
Depth Estimation

- To improve the result
 - Instead of saving only the maximum we can store all the obtained values
 - One function per pixel
-
- This forms a matching cost volume
 - In which we can perform an optimization called graph-cut

Keep all values {
One pixel
One pixel
One pixel

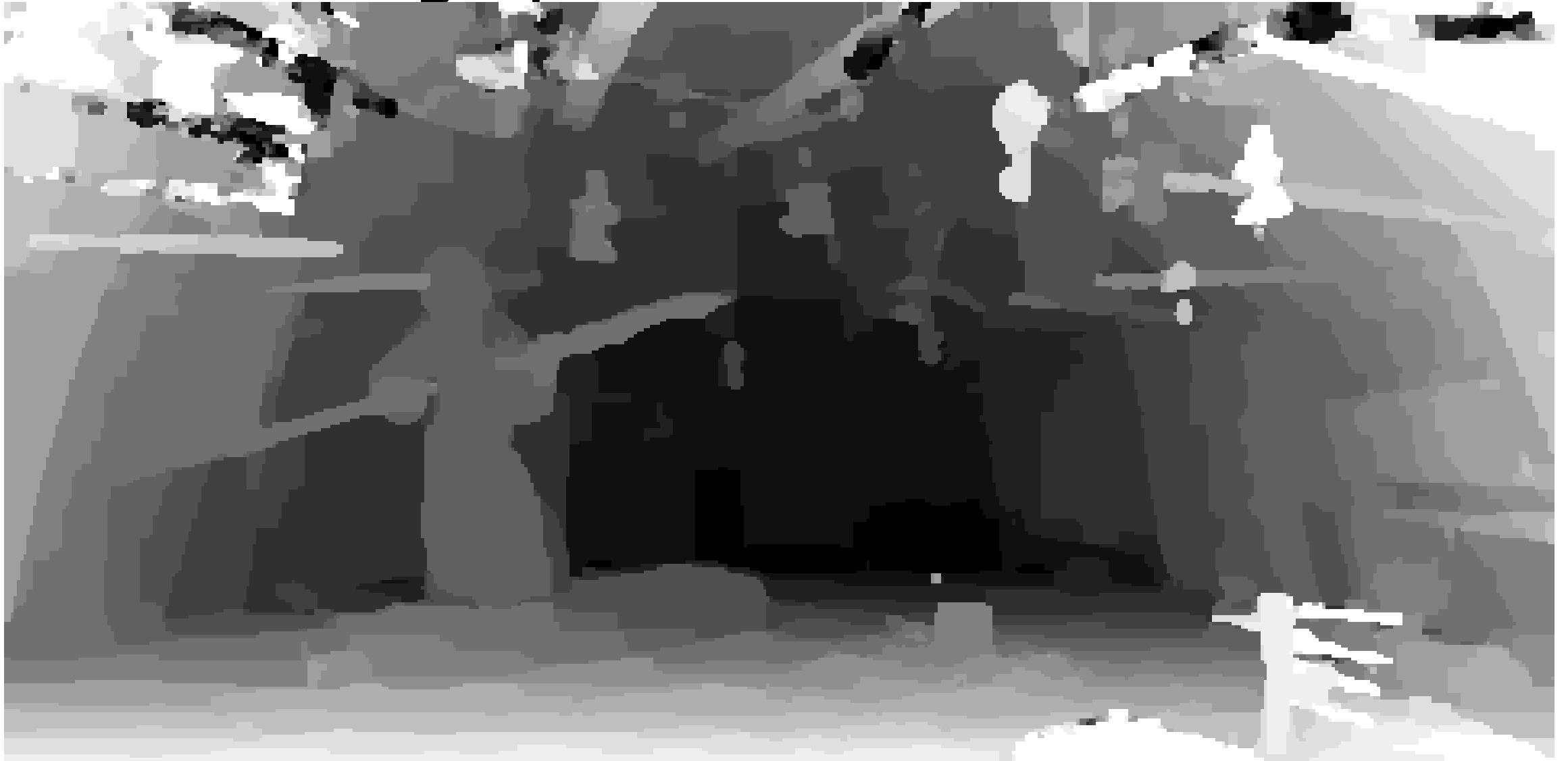


Depth Estimation





Evolution of graph-cut– 480x270 – 1FPS



Evolution of graph-cut– 1920x1080 – 1FPS

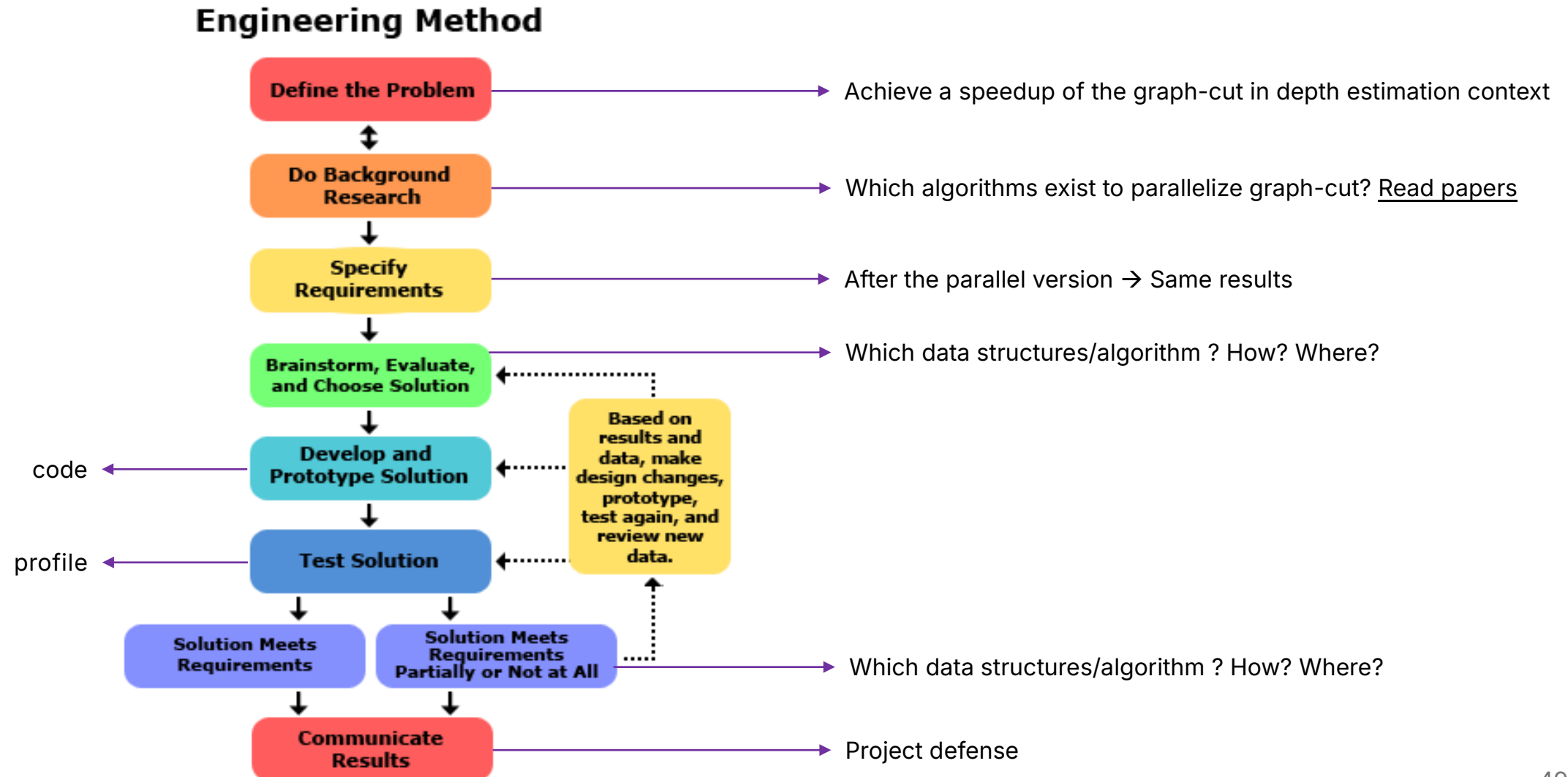


Project

- You will receive a reference code in C++ to do the plane sweeping and the graph-cut
- You need to accelerate the plane sweeping algorithm !
- You don't need to change the graph-cut. It is only used to compare the results
- It is allowed to make algorithmic changes, as long as the end results is visually "very similar"
- Target acceleration = min. one order of magnitude, preferably two orders of magnitude
- You are not only evaluated on the speed-up but also on how you achieved it
 - Put profiling images
 - Explain why some optimizations
 - As discussed in the lectures
- PS: preferably 2 students/group, but 1 is allowed (score +1) & 3 are tolerated (score -1)

Project

- Follow as much as possible the Engineering Design Process (highly correlated to the scientific method)



Ressources

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 - Word Document  - Image  - 3D

List of Resources

	D1.1 – Camera Calibration and Light Field Processing Topic:  File type:  View
	D1.1 – Annex B Topic:  File type:  View
	D1.3 – Multiview and Light field video recordings of real scenes Topic:  File type:  View
	D2.1 – Test conditions and refactorization Topic:  File type:  View